

1.

$$\text{MVT} \Rightarrow \frac{f(2) - f(0)}{2 - 0} = f'(c), \quad c \in (0, 2)$$

$$f(2) - f(0) = 2f'(c)$$

$$f(-x) = -f(x)$$

$$f(-0) = -f(0)$$

$$\Rightarrow 2f(0) = 0 \quad \therefore f(0) = 0$$

$$2 \leq 2f'(c) \leq 10$$

$$f(2) - f(0) \geq 0 \quad \therefore 9 \geq 1$$

2.

$$\lim_{x \rightarrow 0} \frac{3e^{3x} - 3e^{\frac{3}{2}x}}{-3\sin 3x + 4\sin 2x - \sin x}$$

$$\lim_{x \rightarrow 0} \frac{9e^{3x} - \frac{9}{2}e^{\frac{3}{2}x}}{-9\cos 3x + 8\cos 2x - \cos x}$$

$$= \frac{9 - \frac{9}{2}}{-9 + 8 - 1} = -\frac{9}{4}$$

$$24 \frac{1}{4} \times \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 + \cos x} = ??$$

$$\lim_{x \rightarrow 0} \frac{1 - \frac{\sin x}{x}}{1 + \frac{\sin x}{x}} = 1, \quad \left| \frac{\sin x}{x} \right| \leq \left| \frac{1}{x} \right| = \frac{1}{x}$$

3.

$$f'\left(\frac{1}{2}\right) = \frac{1}{f'\left(\frac{1}{2}\right)} = \frac{1}{f'(1)} = \frac{1}{1 + \frac{1}{2}} = \frac{2}{3}$$

$$f' = x^{-1} + \frac{1}{1+x^2} \quad f'(1) = 1 + \frac{1}{2} \quad f'' = -x^{-2} + (1+x^2)^{-2} \cdot 2x$$

$$[f^{-1}(f(x)) = x]' \Rightarrow f^{-1}(f(x)) \cdot f'(x) = 1$$

$$\Rightarrow f''(f(x)) \cdot f'(x) + f^{-1}(f(x)) \cdot f''(x) = 0$$

$$x=1 \Rightarrow f''(f(1)) \cdot f'(1) + f^{-1}(f(1)) \cdot f''(1) = 0$$

$$\Rightarrow f''\left(\frac{1}{2}\right) \cdot \frac{2}{3} - 1 = 0 \quad \therefore f''\left(\frac{1}{2}\right) = \frac{3}{2}$$

4.

$$\lim_{x \rightarrow 3} f(x) = f(3) \text{ 일 } \lim_{x \rightarrow 3} x = 9$$

$$i) x \in \mathbb{R} \quad (\lim_{x \rightarrow 3} x = 9)$$

$$\forall \varepsilon > 0, \quad 0 < |x-3| < \delta_1 \Rightarrow |x^2-9| < \varepsilon \text{ 인 } \delta_1 \text{ 를 찾자}$$

$$\text{원시 } \delta_1 \text{ 를 잡으면 } 2 < x < 4 \quad (\text{참. } x \neq 3)$$

$$\text{반면 } |x^2-9| = |x-3||x+3| < \delta_1 \text{ 역시}$$

$$\delta_1 = \min(1, \frac{\varepsilon}{7})$$

$$ii) x \in \mathbb{R}^c \quad (\lim_{x \rightarrow 3} 3x = 9 \text{ 일 } \lim_{x \rightarrow 3} x = 9)$$

$$0 < |x-3| < \delta_2 \Rightarrow |3x-9| < \varepsilon \text{ 인 } \delta_2 \text{ 를 찾자}$$

$$\text{반면 } |3x-9| = 3|x-3| < 3\delta_2 \text{ 역시 } \delta_2 = \frac{\varepsilon}{3}$$

$$\therefore \delta = \min(\delta_1, \delta_2) \text{ 이면, 위 두 경우 모두 증명된다.}$$

5.

$$f = \begin{cases} x^2 & x \in \mathbb{R} \\ a & x \in \mathbb{R}^c \end{cases}$$

$$\lim_{x \rightarrow a} f(x) = f(a)$$

$$i) a \in \mathbb{R}$$

$$\lim_{x \rightarrow a} f(x) = a^2$$

$$ii) a \in \mathbb{R}^c$$

$$\lim_{x \rightarrow a} f(x) = a$$

$$\lim_{x \rightarrow a} f(x) = \begin{cases} \lim_{x \rightarrow a} x^2 = a^2 \\ \lim_{x \rightarrow a} a \end{cases}$$

$$\lim_{x \rightarrow a} f(x) = \begin{cases} \lim_{x \rightarrow a} x^2 = a^2 \\ \lim_{x \rightarrow a} a \end{cases}$$

$$\text{따라서 } a^2 = a \text{ 일 } a = 1 \text{ (} \because a > 0 \text{)}$$

$$\rightarrow \lim_{x \rightarrow 1} f(x) = f(1) = 1^2$$

$$i) x \in \mathbb{Q} (x \rightarrow 1)$$

$$\lim_{x \rightarrow 1} x^2 = 1$$

$\forall \varepsilon > 0$ 에 대해

$$0 < |x-1| < \delta \Rightarrow |x^2-1| < \varepsilon$$

인 δ 를 찾자.

우선 $\delta=1$ 로 잡으면 $0 < \delta < 2$ 가 된다.

$$\text{반면 } |x^2-1| = |x||x-1| < 3\delta$$

$$\therefore \delta = \min(1, \frac{\varepsilon}{3})$$

$\therefore \lim_{x \rightarrow a} f(x) = f(a)$ 이라 할 수 있다.

$$ii) x \in \mathbb{Q}^c (x \rightarrow 1)$$

$$\lim_{x \rightarrow 1} 1 = 1$$

(상한하한법칙)

$\forall \varepsilon > 0$ 에 대해

$$b < |x-1| < \delta \Rightarrow |1-1| < \varepsilon$$

이므로 모든 x 에 대해 성립한다.